

Project 2: Power System Simulator

Introduction

Project 2 builds on Project 1s structure by developing a fully functional power system analysis simulator. This new simulator can solve both power flow problems and symmetrical fault studies. Centered around the Newton-Raphson iterative method, the program converges on bus voltage magnitudes and phases. The system is built around a class structure, where each element of the power system, such as buses, generators, loads, transmission lines, and transformers, are represented by its own class. This design makes the simulator easy to expand and easy to follow.

How It Works

The simulator is built around a central Circuit class that holds all network components. Each component, buses, generators, loads, transmission lines, and transformers, is its own class and handles its own calculations. Once the network is built, the Circuit assembles the Ybus matrix from each element's primitive admittance. From there, the Newton-Raphson solver iterates by computing the power mismatch, building the Jacobian, and updating bus voltages and angles until convergence. For fault studies, generator subtransient reactances are added as shunts to the Ybus, which is then inverted to get the Zbus and used to calculate fault current and post-fault bus voltages.

How to Use the Simulator

The simulator runs from the main.py. When launched, the user is prompted to name their circuit and is then presented with a numbered menu correlating to different action, to build the network step by step.

Step 1: Add Buses

Select option 1 from the menu. For each bus, enter a name, a nominal kV rating, and a bus type (Slack, PV, or PQ). For Slack and PV buses, you will also be asked to enter a voltage setpoint in per unit. Press Enter to finish adding buses.

Step 2: Add Network Elements

Use options 2 and 3 to add transmission lines and transformers. For each element, you will be asked to specify the two buses it connects, along with its per-unit R and X values. Transmission lines also require shunt G and B values. All values should be entered in per unit on the system base (100 MVA).

Step 3: Add Generators and Loads

Use options 4 and 5 to add generators and loads. Generators need a connected bus, a voltage setpoint, an MW output setpoint, and a subtransient reactance X in per unit. Loads only require a connected bus and MW and Mvar values.

Step 4: Run Analysis

Select option 6 to run the analysis. You will be asked to choose between powerflow mode and fault mode.

- **Power Flow:** The Newton-Raphson solver runs automatically. After convergence, the final voltage magnitude and angle for each bus, along with the real and reactive power injections, are printed to the console.
- **Fault Study:** You will be asked to select the faulted bus and enter the pre-fault voltage. The program will then output the fault impedance Z_{nn} , the subtransient fault current I_F , and the post-fault voltage at every bus in the system.

Notes

- The system base MVA is set to 100 MVA by default in Settings.py.
- All power and impedance values entered should be in per unit on this base.
- The convergence tolerance defaults to 0.001 pu and maximum iterations to 50, both adjustable in Settings.py.
- Generators attached to Slack or PV buses automatically set that bus's voltage to the generator setpoint.
- For fault analysis, generator subtransient reactance must be set when adding the generator.